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Quality of Service Evaluation and Forecast for EV Charging Based on Real-World Data

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Abstract— **In line with the global push towards smart cities, the world is increasingly adopting Electric Vehicles (EVs). This increased EV proliferation is putting the Public Charging Infrastructure (PCI) under a large strain. To this end, this work presents a data-driven analysis of the Quality of Service (QoS) on the current EV PCI. This work presents a comprehensive set of metrics that are developed to evaluate the QoS at the current PCI in Quebec, Canada. The analysis is performed on a real dataset covering 5 full years of over 7,000 EV Charging Stations (EVCSs) in Quebec. This data is then used to create a forecast model for predicting future EV charging requests and assessing their impact on the QOS at the current PCI deployment levels. The developed metrics and forecast model are used to recommend new EVCS deployment sites to guarantee acceptable QoS levels in the future based on the current trends in EV adoption.**

Index Terms **- Electric Vehicles, Public Charging Infrastructure, Demand Forecast.**

I. INTRODUCTION

The increasing need for reducing GreenHouse Gas (GHG) emissions and the push for smart cities have promoted the use of Electric Vehicles (EVs) [1]. EVs are becoming an increasingly popular option for the reduction of the emissions of the transportation sector which contributes to a large portion of our total emissions, i.e, 27% of total emissions in Canada [2].

The main benefit of EVs is their reliance on electricity which is a cleaner source of energy than gasoline even in the presence of power plants that still rely on the burning of fossil fuels. In British Colombia and Quebec, where 90 to 99% of electricity is generated by clean hydroelectric power, EVs offer a 99% emission reduction even when the emissions in the manufacturing process of EVs are accounted for [3].

To successfully take advantage of the benefits offered by EVs, 195 countries signed the Paris Climate agreement to reduce GHG emissions and set self-imposed targets to reach a 30% EV market share by 2030. To this end, governments around the world offer incentives for EV purchases ranging from rebates on purchases to road tax exemptions. Further government initiatives include plans to expand the charging infrastructure. The US government has dedicated 8 billion USD to establish a sufficient charging infrastructure to support the anticipated growth in EV numbers. The Canadian Government has also dedicated 900 million CAD to establish a network of additional 50,000 EVCSs. The technological advancements and larger production volumes coupled with reduced EV prices have improved public opinion on EVs.

As such, we are witnessing exponential growth in EV numbers on the road. By the end of 2019, there were 7.2 million EVs on the road [4]. The exponential growth is demonstrated by the record EV sales of 3.2 million in 2020 [4], 6.7 million in 2021 [4], and 10 million in 2022. The Quebec EV market has also grown to 160,000 EVs by the end of 2022. In line with the global and Canadian vision of achieving 100% zero-emission vehicle sales, Quebec has set a target of 1 million EVs by 2030 [5] and will ban new gasoline car sales in 2035 [6].

This raises a serious question of sufficient EV Charging Station (EVCS) deployment to meet the rising charging demand and maintain the QoS for EV drivers. The rapid EV adoption is accompanied by great concerns for EV users namely, range anxiety and the frequent need to recharge EVs. As more people switch to EVs, the demand for public EVCSs will continue to grow. This stresses the need to expand the Public Charging Infrastructure (PCI) and ensure that EVCSs are readily available to EV drivers to support the continued growth of the EV market. EVCS operators and manufacturers are trying to address this issue by deploying more EVCSs with a higher charging rate (Level 3 chargers) and by adopting smart scheduling techniques [7] [8]. The main aim of these plans is to decrease the EV drivers' waiting time and foster trust in the PCI. However, the current measures are not enough to address the issue of waiting time due to the lack of practical reservation systems or a check-in system that keeps track of waiting time. Consequently, due to the lack of visibility by the operators for the waiting times, operators tend to approximate the waiting time based on assumptions, such as constant charging rate, constant battery capacity, etc.

Currently, the International Energy Agency (IEA) and several governments use the ratios of EVs to EVCSs as a metric to determine the adequacy of the charging infrastructure deployment. The global average is around 10 EVs per public EVCS [4]. While this is considered an acceptable ratio we mention that some countries especially in North America have ratios as high as 20 EVs per EVCS. In Quebec for instance, this ratio has been constantly increasing from 5 EVs per EVCS in 2018 to 17 EVs per EVCS by the end of 2022. This demonstrates how the PCI is in dire need of expansion to catch up with the increasing rate of EV adoption.

These metrics, however, can be deceiving and do not provide a complete picture. First, EV adoption levels differ from one city/area to another within the same country or province. Additionally, the EVCS deployment differs significantly based on location. Finally, the charging demands vary considerably depending on the nature of the region. For instance, while deployment levels might appear satisfactory in a touristic area, it is important to keep in mind that the majority of EVCS users in touristic areas will be from out of town. These users will not be accounted for by using the aforementioned ratios. To this end, it is critical to develop proper mechanisms and metrics that can properly evaluate the QoS of EVCSs per location.

Consequently, in this work, we propose a methodology that provides operators with an extensive list of QoS evaluation metrics. Our approach is data-driven based on the real-world dataset provided by our partner Hydro-Quebec, a major power-grid utility in North America. The proposed operational metrics include EVCS occupancy ratios, blocking probability, and charging behavior trends. Such metrics are crucial elements that can be used by operators to take informed decisions for optimizing deployment and enhancing the QoS drastically by devising scheduling algorithms and resizing the PCI network to achieve the smart cities of the future. Additionally, we devise a long-term forecast model to predict the future charging demand per EVCS site. Based on the identified gap in proper QoS evaluation metrics in the literature, and the lack of proper long-term EV charging demand forecast tools, we summarize the contributions of this work as follows:

- To the best of our knowledge, we are the first to devise a comprehensive data-driven framework to assess the QoS of EVCSs. We develop a set of extensive metrics that aids the PCI operator gain a deeper understanding of their charging infrastructure utilization. These metrics also provide extended knowledge that could aid operators to optimize the PCI deployment, sizing, and scheduling to attain the global goal of attaining smart cities.
- We developed a Machine Learning (ML) algorithm for long-term EV charging demand forecast based on historical data. By taking into consideration multiple factors in addition to the number of requests, our model was able to accurately predict up to 1 year of future EV charging demand.

The rest of the paper is organized as follows. Section II Discusses the related works and their gaps. Section III discusses the devised metrics as well as the forecast model. Finally, Section IV presents the results of our evaluation and forecast on several selected sites. The paper is concluded in Section V.

II. BACKGROUND AND RELATED WORK

The problem of forecasting EV charging demand has witnessed a lot of attention in recent years. Nonetheless, most of these studies rely on simulated data or small datasets, unlike our study which relies on a dataset for 7,000 EVCSs over 5 full years for the entire province of Quebec. This dataset is used to develop an entire set of metrics on which we base our evaluation and forecast.

The work performed in [9] is a survey of PCI planning studies that focuses on their different objectives and gaps. The authors found that the majority of such studies focus on theoretical approaches and simulations instead of using real data. Furthermore, these studies focus on infrastructure planning from a mobility or power distribution network perspective with a few of them using a combined approach. More importantly, most studies focus on optimizing the installation cost with only a few studies focusing on the quality of service for the EVs/owners.

The authors of [10] provide a multistage planning model for the deployment of EVCSs in distribution grids. The model tries to balance the investment cost and peak demand of the distribution grid. While the waiting time of EVs at the EVCSs is also considered, it is not the main objective of their model. Additionally, the model is not formulated as an optimization problem which hinders its ability to efficiently optimize the three different objectives simultaneously. Furthermore, the model is built on mobility data without any data on the performance and utilization of the currently deployed PCI.

The authors of [11] implemented a sophisticated model for estimating charging demand in a geographical area. To this end, they utilize real data that is obtained by crawling ChargePoint and GoogleMaps for any information that can be obtained on the utilization of the charging infrastructure. This data is then fed into a Modified geographical PageRank to describe EV drivers' willingness to utilize each EVCS. This data is then fed into a regression model to map it into an actual estimation of the charging demand. This charging demand is then fed into a capacitated maximal coverage location problem to optimize the investment cost needed to assure the maximum geographical coverage of the charging infrastructure. Their model, however, relies on partial data fetched from the internet with no guarantee of complete knowledge of the existing PCI or the total demand per existing EVCS. Furthermore, their method offers no in-depth analysis of the quality of service at each charging site.

A short-term prediction model, up to 4 hours, was devised in [12] using a series of Hybrid LSTM models. This model accurately predicts EVCS occupancy based on a real dataset to create a realistic model that can be incorporated into a smart charging policy. Although the model achieves 99.99% accuracy, it is meant for very short-term forecasts and cannot be relied upon for long-term PCI planning.

Additionally, the authors of [13] created an ARIMA model to predict the day-ahead charging demand in a small power grid. The authors found that the most accurate prediction can be made if the traditional electric demand excluding EV load and EV charging demand are forecasted independently of each other. This model, however, is not meant to study the QoS at the individual EVCSs. Additionally, a day-ahead prediction cannot be utilized for long-term infrastructure planning.

Finally, a comparison between different Deep Learning (DL) models was performed in [14] for the prediction of short-term charging demand. All the models were trained based on a real dataset. While the multivariate LSTM and multivariate gated recurrent units achieved an accuracy of 96%, these models were only optimized to predict 72-hour periods.

III. METHODOLOGY AND ATTACK MODEL

The following section discusses the metrics we devised to evaluate the performance of the EVCS infrastructure at the current deployment levels. A forecast model is then developed to predict the future EV charging demand. This predicted load will in turn be used as input to a custom-built simulator to analyze the behavior of the EVCS based on the added charging demand. These metrics and models can be used by utilities and operators to determine the charging sites that will experience a drop in QoS below acceptable thresholds. This in turn can be used to determine the locations for installing new EVCSs.

A. Dataset Analysis

To the best of our knowledge, this work is the only study that relies on a complete and comprehensive dataset provided by an EVCS operator. We obtained this dataset as part of a research collaboration and legal agreement, with Hydro-Quebec, which owns and operates, through their subsidiary Circuit Electrique, the largest public EVCS network in Quebec.

This complete dataset contains the historical data starting from January 1, 2018, till December 31, 2022. It contains the records of over 6 million charging sessions on 7,454 EVCSs distributed over 3,878 different sites. These records contain the start time of each charging session as well as its duration. The dataset, however, does not contain any information about the arrival time of the EV or the time it spent waiting. A site is defined as a small geographic location (i.e. parking lot) that contains multiple EVCSs. Using this dataset we create realistic metrics for the evaluation of the QoS on each EVCS independently or in conjunction with nearby EVCSs. To clarify, we consider that multiple EVCSs installed at the same location are part of the same charging site. Thus these EVCSs share a common waiting queue which will impact the formulation of the metrics describing this site.

B. Performance Metrics

As discussed above, we rely on our real-world dataset and develop an extensive set of performance metrics. These metrics are later used to evaluate the QoS of several charging sites.

1) **Number of Charging Requests (N_R):** This is the main metric that can be extracted directly from the dataset. N_R is computed based on the number of unique charging requests on an EVCS during a given period. We disregard any charging session that lasts only for a few seconds.

2) **Charging Site Utilization (U):** This is one of the most important metrics that is used to determine if a site is being underutilized or over-utilized. By studying the evolution of this trend during the different times of day, an operator can determine the exact demand for charging at each site during a given time of day. U also varies between different seasons and months of the year depending on the changing EV charging needs and user habits. This gives the operator a clearer understanding of the actual state of demand on the EVCS that cannot be provided by N_R. U is calculated based on the actual utilization EVCSs per site according to Equation (1) where T is the total number of minutes per period, k is the number of occupied EVCSs, n is the number of EVCSs per site, and i is the index of each minute in the studied period.

$$U = \frac{1}{T}\sum_i \frac{k_i}{n} \quad (1)$$

3) **Occupancy (Ω):** This is the probability that at least 1 EVCS is occupied per site. This metric is used to study the probability that EVs use the site under consideration. Ω is expressed as $P(k_i \geq 1)$ and calculated using Equation (2).

$$\Omega = \frac{1}{T}\sum_i Z_i \text{ where } Z_i = \begin{cases} 0 \; if \; k_i = 0 \\ 1 \; if \; k_i \; \geq 1 \end{cases} \quad (2)$$

4) **Blocking Probability per site (P_B):** This is the ratio of the time all EVCSs were busy at a site over the total time. This is an extremely important metric quantifying the probability of an EV arriving at a given site and finding all EVCSs occupied. P_B demonstrates to the operator the trend of people being denied immediate service due to a lack of sufficient EVCSs per site. It is considered problematic when this number exceeds a certain threshold. P_B can be calculated using Equation (3).

$$P_B = \frac{1}{T}\sum_i X_i \; where \; X_i = \begin{cases} 0 \; if \; k_i \geq n \\ 1 \; if \; k_i \; = n \end{cases} \quad (3)$$

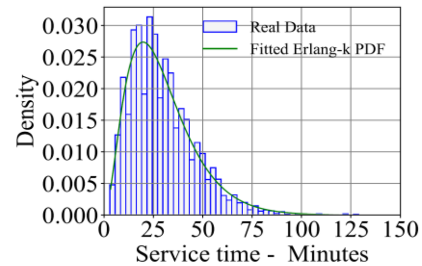


Figure 1. Sample service time data following an Erlang-K distribution.

5) **Number of EVs that waited to receive service (N_{WE}):** This is an estimate of the number of EVs that waited at a given site. An EV is considered to have been waiting if it starts charging within t minutes after a charging session ends at a blocked site. The value of t is empirically estimated by the operator based on the availability of parking spots in the vicinity of the charging site and their distance from the EVCS.

6) **Average Waiting Time (WT_{Av}):** is calculated based on a custom-built simulator we developed that models each site as an M/G/K queue. This simulator is then used to simulate the arrival time, charging time, and the time each EV spends in the queue. The average time spent by the EVs in the queue is considered the average waiting time at this site.

7) **Site QoS Evaluation:** Based on the metrics presented above, we perform the analysis of QoS analysis of each charging site. While each metric on its own generates important insight into the operation of the charging site in a given period of interest, it should not be used to evaluate the QoS for the entire month/year as a single period. Instead, we propose the evaluation of the devised metrics per day to evaluate the QoS during weekdays, weekend days, work days, and holidays independently. After these metrics have been evaluated per day, we examine the number of days during which each metric was in a certain range or exceeded a given threshold.

The advantage of this approach lies in not diluting the impact of "special days" where the QoS was in a certain range of interest to the operator, by aggregating it with the other normal operation days. In touristic areas, for instance, the overall QoS might be considered good if the values of the above metrics for an entire year or month were relatively low when in reality the days of interest to the operator are holidays and weekends. To this end, it becomes important not only to study the value of these metrics but rather the number of days during which the values of these metrics fall within different ranges.

C. Simulation Framework

Next, we developed a fully customized simulation framework following an M/G/K queuing model. The M, G, and K parameters of this model are tuned based on the records of each site extracted from our dataset. The arrival rate, M, of EVs at the charging site is modeled as a Markovian process modulated by a Poisson process. The inter-arrival time of this Poisson process is calculated based on the number of requests extracted from the dataset. Additionally, the service time distribution (G) is derived from the 6 million charging records in the Hydro-Quebec Dataset. After examining the distribution of the charging time per site it is found to follow an Erlang-K distribution. Figure 1 demonstrates a sample service time distribution one of the sites and demonstrates the fitted Erlang-K distribution. Finally, the number of servers per queue

(EVCSs per site) is represented by K. The main objective of the developed simulator is to simulate the EV arrival at the charging site based on the real-world data we have. This simulation allows us to reliably obtain the missing information required to calculate the QoS metrics such as average waiting time. Additionally, this simulator will be used to incorporate the future EV arrival and charging based on the forecasted future requests for the year ahead. These results will also be used to calculate the QoS metrics during the forecasted period.

D. Filling the COVID GAP

Due to the exceptional circumstances caused by the COVID-19 pandemic, the EVCS infrastructure experienced a sharp drop in demand. During this period, the number of charging requests on all the sites experienced a decline due to the lockdowns, curfews, border closure, and remote work policies implemented across Quebec and the world. Thus, the 18 months interval between January 2020 and June 2021 (referred to as Gap hereinafter) needs to be addressed to represent the normal charging demand that would have been experienced if the pandemic never occurred. Hence, we address this issue using the following procedure:

- After obtaining the entire 5-year records of a given site, the requests during the Gap are discarded.
- Due to the randomness of the daily charging requests, a 4-week moving average is calculated by taking the average of the 4 weeks centered at each specific day of our 5-year period using Equation (4) where d is the date of each specific day we are computing the moving average at and N_{R_j} is the number of requests during the j[th] day.

$$MAV = \frac{1}{29}\sum_{j=d-14}^{d+14} N_{R_j} \tag{4}$$

- Figure 2 demonstrates how the moving average captures the seasonal and annual trends in the number of charging requests while smoothing out the daily randomness.
- We then use this average in the non-COVID months to interpolate the weighted moving average during the Gap.
- The next step is creating a mathematical distribution to capture the relative relation between the moving average and the daily number of charging requests.
- First, the difference between the actual number of charging requests and the value of the moving average is calculated for each day and then normalized by the value of the moving average to obtain the relative difference.
- This relative difference is then collected for the entire 42-month period before and after the Gap. This data is then used to fit a mathematical distribution representing the relative difference between the actual number of requests and the weighted moving average.
- Finally, by using the value of the moving average on each day and a randomly generated relative difference following the distribution obtained above we compute the number of requests for all the days of the 18-month Gap.

Figure 2 demonstrates how this method allowed us to fill the Gap while preserving all seasonal and annual trends.

E. Forecast Model

1) **Data Pre-Processing:** The first step of the pre-processing is the extraction of the charging records chronologically to create a time series instead of scattered requests throughout the dataset. The main feature of this time series is the timestamp that indicates the season and time of year. Additionally, as part of the feature engineering process, a set of external features is also fed into the model. These features are external factors that impact the charging demand. These factors are the number of EVCSs in the geographic area, the number of EVs registered in that area, and the number of EVs registered in the province The features are then encoded using the mean encoding method [15]. This encoding technique provides a logical correlation between the feature and the label, i.e. the number of charging requests. During the training phase, the model calculates the relationship between the predicted value and mean encoding of the features instead of the actual values of these features.

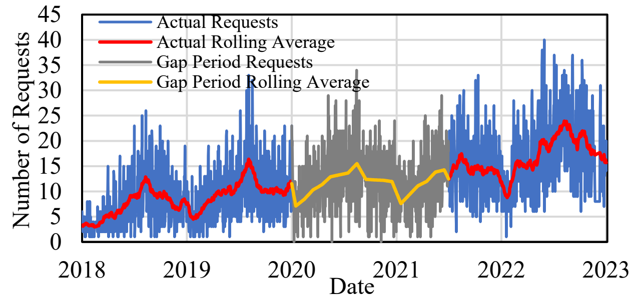


Figure 2. Daily charging requests including filling the Gap

2) **Forecast Model:** The EV charging demand is expected to grow at an increasing pace in the coming years. This stresses the importance of creating a long-term forecast model that can accurately predict the EV charging load in terms of charging requests covering a forecasted period equal to an entire year while considering the load's seasonality. To this end, we develop a Seasonal Auto-Regressive Integrated Moving Average with eXogenous factors (SARIMAX) [16] time-series-based model that is a statistical learning model capable of predicting the future charging requests demands based on provided historical real-world data.

SARIMAX can accurately capture the seasonal changes in the input data to accurately represent this seasonality for the 1-year forecast. SARIMAX is also selected due to its ability to integrate multiple external variables and deduce their impact on the behavior of the charging demand. These features can ensure added accuracy of the model.

The structure of the SARIMAX model can now be tuned to fit the presented historical data. SARIMAX is defined by the p, d, and q parameters. These parameters represent the number of autoregressive terms, the order of differentiation, and the order of moving averages respectively. This is then followed by a calculation of seasonal variation to achieve a seasonal auto-regressive (SAR) function in addition to the non-seasonal auto-regressive (AR) function [17]. Additionally, the set of exogenous variables is presented to the model as "exog" parameters. By properly tuning the SARIMAX model, it can provide accurate predictions and generate forecasted results based on the detected patterns and the exogenous variables. The recorded daily number of charging sessions per site and their corresponding exogenous variables are aggregated weekly to overcome the randomness of the daily number of requests. This kind of aggregation is acceptable and rather preferred for applications such as ours that aim at evaluating the evolution of the quality of service in the future. The reliability of the developed model is evaluated based on accuracy metrics such as Root Mean Square Error (RMSE).

IV. RESULTS

This section presents an evaluation of the performance of 5 selected charging sites based on the above-presented metrics in Section III. We also demonstrate our SARIMAX forecast model on one site and demonstrate how the average waiting time evolves between 2021 and 2022 as well as the predicted waiting time for 2023.

A. Occupancy, Utilization, and Blocking Probability

For demonstration, 5 diverse charging site were selected and evaluated. These 5 sites are 2 cities, 1 touristic town, 1 suburban area, and 1 site next to a highway. The charging sites in the Touristic Town, Highway, and City 2 all have 2 EVCSs. However, the sites in City 1 and the Suburban area have 1 EVCS. Table I presents a summary of the number of days where the utilization and occupancy at each site exceeded 30%. We present these 2 metrics for both 2021 and 2022 to demonstrate the evolution of PCI utilization. We note that when a given site has only 1 EVCS, the values of Ω and U will be equal.

Table I: Number of days Ω and U were exceeded 30%

Metric / Site	# Days with $\Omega > 30\%$		# Days with U > 30%	
	2021	**2022**	**2021**	**2022**
Touristic Town	110	263	27	123
Highway	5	6	0	1
Suburb	52	113	52	113
City 1	3	10	3	10
City 2	107	232	12	38

We highlight that on all these 5 sites both Ω and U witnessed an increase. This is especially true for the Touristic Town, the Suburban area, and City 2 where the number of days Ω exceeded 30% doubled. Another insight that can be extracted from Table I is that the charging site next to the Highway is underutilized where the utilization, U, exceeded 30% on zero days and 1 day in 2021 and 2022 respectively. This could indicate the user's tendency to charge at their starting point or the destination rather than spending time charging on route.

Another important metric we studied at these 5 sites is the blocking probability, P_B. Due to the sensitivity of this metric, we present in Table II in 10% intervals. A P_B of zero means that this site was not blocked on a given day. After inspecting the data in Table II, it is evident that the charging site next to the Highway had a chance of not being blocked for 295 days and 256 days in 2021 and 2022 respectively. This confirms our previous observation that EV drivers were less likely to use this specific site. If we examine all 5 sites we notice that the number of days P_B.was zero decreased between 2021 and 2022. This is in line with the data presented in Table I since Table 1 demonstrated how the occupancy and utilization of these sites increased in 2022. The most shocking result, however, is the blocking probability in the suburban area. While in 2021 this site had a P_B> 20% for 106 days, this number almost doubled to reach 206 days in 2022. This means that an EV arriving at this site had a 20% chance of not finding an available EVCS on

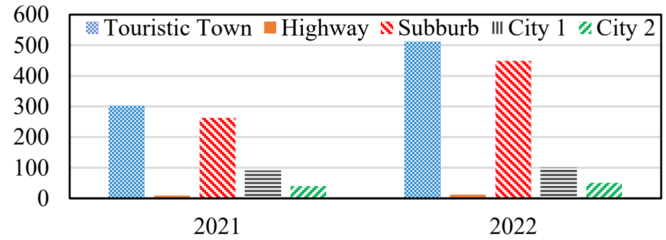


Figure 3. Evolution of N_{WE} on the 5 sites between 2021 and 2022

206 days out of the 365 days that year. This is a clear indication that an additional EVCS needs to be added at this site or close by in the same region. While the charging site in City 2 also had a low chance of having P_B= 0, its performance was still acceptable in 2022 as most of the days had a P_B between 0% and 10%. These results reflect the number of EVs that had to wait before receiving service as demonstrated below.

Table II: Blocking probability evolution from 2021 to 2022

P_B	0%		0%-10%		10%-20%		>20%	
Year	**2021**	**2022**	**2021**	**2022**	**2021**	**2022**	**2021**	**2022**
Touristic Town	179	45	132	161	43	104	11	55
Highway	295	256	69	108	1	1	0	0
Suburb	34	6	109	48	106	105	116	206
City 1	20	10	215	155	104	149	26	51
City 2	66	43	268	246	29	69	2	7

B. Delayed EVs

Perhaps the most important metric is the N_{WE} that represents the number of EVs that had to wait at a given charging site because the site was blocked. As a result, there is a direct relationship between the P_B and the N_{WE}. This is demonstrated in Figure 3 which represents the evolution of N_{WE} between 2021 and 2022 at the 5 sites. From this distribution, we can deduce that the number of EVs that had to wait to receive service at the charging site in the Touristic town increased by 67% which means the service will quickly deteriorate at this site in the coming year if the PCI in the area is not properly expanded. The results at the Highway charging site are consistent with the above tables since only 9 and 12 EVs had to wait for the entire year of 2021 and 2022 respectively.

C. SARIMAX Model Testing and Forecast

Based on the Box-Jenkins approach, the behavior of the time series data is used to define the degree of differentiation required to attain constant mean, variance, and covariance. The differentiation order is selected based on the partial/auto-correlation function. This step is then followed by executing Augmented-Dickey-Fuller tests to verify the existence of non-stationary conditions. The model then learns the time-series behavior by examining the residuals at the end of each estimation stage. Finally, the model is trained on the data of the Touristic area for the period between 2018 and 2021 and tested against the 2022 data. The model is continuously tuned until the RMSE drops to 15.81. The model accuracy is demonstrated in Figure 4 where the forecast very closely follows the actual time-series. The shaded region also indicates a 99% confidence interval such that the forecasted number of requests will fall within this region with a 99% confidence. This model is now used to forecast the demand for the entire year of 2023 as demonstrated in Figure 5.

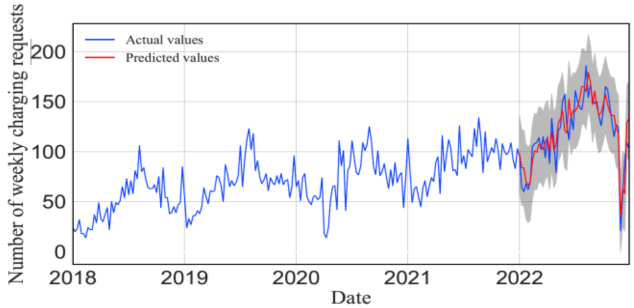


Figure 4. Validation of the SARIMAX model accuracy on the 2022 records

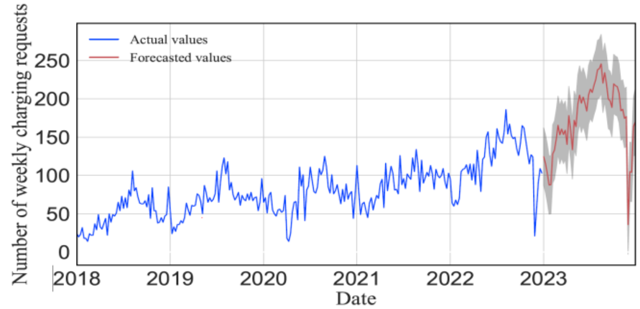
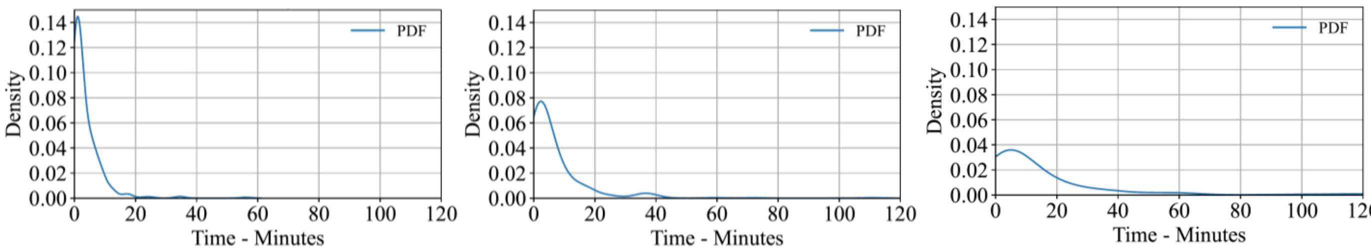


Figure 5. SARIMAX model forecast for the 2023 weekly charging records



(a) (b) (c)

Figure 6. Evolution of the Average Waiting Time (WT_{Av}) distribution in (a) 2021, (b) 2022, and (c) 2023.

D. *Waiting Time Evolution*

After forecasting the number of requests for 2023 we utilize our M/G/K simulator to study the evolution of the waiting time. To this end, we run our simulator for 2021 (including the filled Gap), 2022, and the forecasted 2023. We first extract the daily number of requests from the forecasted weekly value. Then we use this value to determine the Poisson arrival rate of EVs to the charging site. Then the simulation is performed for the arrival rates of the 365 days of each year. The simulator is also set up to simulate the arrival of 25,000 EVs at each given arrival rate to guarantee the convergence of the simulation results as well as reduce the impact of any outliers.

The results of the three years' simulations are presented in Figure 6. This figure demonstrates how the probability of low waiting times is decreasing while the probability of experiencing high waiting times is increasing. This is demonstrated by the bulkier tale of the distribution and the more frequent WT_{Av} values above 60 minutes. To put things into perspective, the probability of experiencing zero waiting time is 20.4% and 9.3% in 2021 and 2022 respectively. This value drops to 3.8% in 2023. Additionally, the average waiting time in 2021 was 3.49 minutes and rises to 6.73 minutes and 14.68 minutes in 2022 and 2023 respectively. Additionally, 99% of the anticipated waiting times are within 30 minutes, 40 minutes, and 110 minutes in 2021, 2022, and 2023 respectively.

V. CONCLUSION

This is the first data driven study that develops a comprehensive set of metrics to evaluate the performance of the PCI of the smart cities of the future. The herein-developed forecast model also accurately predicts the evolution of charging requests at a given site. We analyzed the performance of 5 representative sites and examined the evolution of the occupancy, utilization, blocking probability, and the number of waited EVs. We also used to the custom-built M/G/K simulator to estimate the average waiting time in 2021 and 2022 as a baseline to compare with the expected waiting time based on the 2023 forecasted charging demand.

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